
purenetwork

Name

purenetwork, purenetwork-create, purenetwork-delete, purenetwork-disable, purenetwork-enable, purenetwork-list, purenetwork-setattr — manages the network interfaces used to connect a FlashArray system to a network

purenetwork-ping, purenetwork-trace — pings remote destinations and traces routes on the network

Synopsis

`purenetwork create vif [--address ADDRESS] --subnet SUBNET INTERFACE`

`purenetwork delete INTERFACE . . .`

`purenetwork disable INTERFACE`

`purenetwork enable INTERFACE`

`purenetwork list [ --cli | --csv | --nvp ] [--notitle] [--page] [--raw] [--service SERVICE] [--subnet SUBNET] --vlan VLAN`

`purenetwork ping [--count COUNT] [--interface INTERFACE] [--no-hostname] [--packet-size PACKET-SIZE] [--user-to-user-latency] DEST`

`purenetwork setattr [--address ADDRESS] [--gateway GATEWAY] [--mtu MTU-SIZE] [--netmask NETMASK] [ --subinterfacelist INTERFACELIST | --addsubinterfacelist INTERFACELIST | --remsubinterfacelist INTERFACELIST ] [--subnet SUBNET] INTERFACE`

`purenetwork trace [--dont-fragment] [--interface] [--method METHOD] [--mtu] [--no-hostname] [--port PORT] [--source SOURCE] DEST`

Arguments

DEST

Internet Protocol (IP) address or full hostname of a ping target or of a remote computer to which the network route is to be determined.

INTERFACE

Network interface name.

Ethernet interface names are in the form CTx.ETHy, where x denotes the controller (0 or 1) and y denotes the interface (0 or 1).

App interface names are in the form APP.datay and APP.mgmtly, where APP denotes the app name and y denotes the interface (0 or 1).

Bond interface names are comprised of alphanumeric characters.

VLAN interface names include a VLAN ID number, which is appended to the name of the physical interface.

In CLI commands, interface names are case-insensitive. For example, CT0.ETH0, Ct0.Eth0, CT0.eth0, and ct0.eth0 refer to the same Ethernet interface name. Likewise, bond1, Bond1, and BOND1 refer to the same bond interface name.

Options

-h | --help

Can be used with any command or subcommand to display a brief syntax description.

--address ADDRESS

IP address to be associated with the specified Ethernet interface.

For IPv4, enter the address in CIDR notation ddd.ddd.ddd.ddd/dd. For example, 10.20.20.210/24. Alternatively, specify the address ddd.ddd.ddd.ddd with a netmask (--netmask).

For IPv6, enter the address and prefix length in the form xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx/xxx. For example, 2001:0db8:85a3::ae26:8a2e:0370:7334/64. Consecutive fields of zeros can be shortened by replacing the zeros with a double colon (: :). Alternatively, specify the address xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx with a prefix length (--netmask).

--addsubinterfacelist (Bond interfaces only)

Comma-separated list of one or more additional slave devices to be added to the specified bond interface. The slave device name should only include the interface name ethX. Do not include the controller name ct0.

Purity//FA disables the interface before it is added as a slave device to a bond interface. If the slave device that you are adding to the bond interface is an administrative interface, it will lose SSH connection and no longer be able to connect to the controller.

--count *COUNT*

Number of Internet Control Message Protocol (ICMP) ping messages to send in sequence.

--dont-fragment

Do not fragment the probe packets. For IPv4 and IPv6, it also sets the Don't Fragment (DF) bit. By default, IP allows the packets to be fragmented when it goes through a segment with a smaller maximum transmission unit (MTU).

--gateway *GATEWAY*

IP address of the gateway through which the specified interface is to communicate with the network. For IPv4, specify the gateway IP address in the form *ddd.ddd.ddd.ddd*. For IPv6, specify the gateway IP address in the form *xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx*. When specifying an IPv6 address, consecutive fields of zeros can be shortened by replacing the zeros with a double colon (*::*).

To remove the gateway specification, set to a null value (*""*).

--interface *INTERFACE*

For **purenetwork ping**, IP address or interface name of the current controller. If the interface is an address, the source address is set to the specified interface address.

For **purenetwork trace**, interface name of the current controller.

--method *METHOD*

Method for trace operations. By default, the trace operation uses the UDP method. Trace operations can also be specified as ICMP (*icmp*) or TCP (*tcp*).

--mtu (purenetwork trace only)

Maximum transmission unit (MTU) along the path being traced between two Ethernet hosts.

--mtu *MTU-SIZE* (purenetwork setattr only)

Maximum message transfer unit (packet) size for the interface in bytes. Valid values are integers between 1280 and 9216 (inclusive). Defaults to 1500 if not specified.

--netmask *NETMASK*

Defines the range of IP addresses that make up a group of IP addresses on the same network. For IPv4, if the address is not entered in CIDR notation, enter the subnet mask in the form *ddd.ddd.ddd.ddd*. For example, *255.255.255.0*. For IPv6, if the address entered did not include a prefix length, specify the prefix length. For example, *64*.

--no-hostname

Do not map IP addresses to hostnames when displaying them.

--packet-size *PACKET-SIZE*

The number of data bytes to be sent. The default packet size is 56. For ICMP tracing, the default packet size is combined with 8 bytes of ICMP header data to equal 64 ICMP data bytes.

--port *PORT*

Initial destination port. The default destination port is 33434.

For UDP tracing, specifies the destination port base used by **purenetwork trace**. The destination port number will be incremented by each probe. For ICMP tracing, specifies the initial ICMP sequence value incremented by each probe. For TCP tracing, specifies the destination port to connect.

--remsubinterfacelist (Bond interfaces only)

Comma-separated list of one or more slave devices to be removed from the specified bond interface.

--service

Lists only the interfaces configured with the specified service. Supported services include (case-sensitive) app, iscsi, management, nvme-roce, and replication.

--subinterfacelist (Bond interfaces only)

Comma-separated list of one or more slave devices to be added to the specified bond interface. In the **purenetwork setattr** command, this option replaces the entire list of slave devices that are currently associated with the specified bonding interface.

The slave device name should only include the interface name ethX. Do not include the controller name ct0.

Purity//FA disables the interface before it is added as a slave device to a bond interface. If the slave device that you are adding to the bond interface is an administrative interface, it will lose SSH connection and no longer be able to connect to the controller.

--source *SOURCE*

Alternative source address. The source address must be the address of one of the interfaces. By default, **purenetwork trace** uses the address of the outgoing interface.

--subnet *SUBNET*

For **purenetwork setattr**, name of the subnet which the physical, virtual, bond, or VLAN interface is to be attached. To detach a physical, virtual, or bond interface from a subnet, set to an empty string ("").

For **purenetwork list**, lists only the interfaces that belong to the specified subnet.

--user-to-user-latency

Prints full user-to-user latency. By default, **purenetwork ping** prints network round-trip times.

--vlan

Lists only the interfaces configured with the specified VLAN ID.

Options that control display format:

--cli

Displays output in the form of CLI commands that can be issued to reproduce the current configuration. The `--cli` output is not meaningful when combined with immutable attributes.

--csv

Lists information in comma-separated value (CSV) format. The `--csv` output can be used for scripting purposes and imported into spreadsheet programs.

--notitle

Lists information without column titles.

--nvp

Lists information in name-value pair (NVP) format, in the form `ITEMNAME=VALUE`. Argument names and information items are displayed flush left. The `--nvp` output is designed both for convenient viewing of what might otherwise be wide listings, and for parsing individual items for scripting purposes.

--page

Turns on interactive paging.

--raw

Displays the unformatted version of column titles and data. For example, in the `purearray monitor` output, the unformatted version of column title `us/op (read)` is `usec_per_read_op`. The `--raw` output is used to sort and filter list results.

Description

Manages the interfaces and the network connection attributes of the array.

Each FlashArray controller is equipped with two 1-gigabit Ethernet (1GbE) interfaces that connect to a data center network for array administration. The interfaces are called CTx.ETH0 and CTx.ETH1, where x denotes the array controller number. Physical interface ports are located on controller rear bulkheads, and are labeled ETH1 (left) and ETH0 (right).

The **purenetwork enable** and **purenetwork disable** subcommands respectively enable and disable a network interface.

Take caution when disabling an interface. If you disable an interface through which an administrative session is being conducted, it will lose SSH connection and no longer be able to connect to the controller.

The **purenetwork list** subcommand lists all the network interfaces for all controllers on the array.

App interfaces only appear for the apps that have been installed. For bond interfaces, the list also includes the bond interface slave devices, if any. For interfaces that belong to subnets, the list also includes the name of the subnet to which they belong. Include the **--service** or **--vlan** option to display only the interfaces that are configured with the specified service type or VLAN ID, respectively.

The **purenetwork setattr** subcommand changes the IP address, netmask, gateway, and MTU attributes of the specified physical, virtual, or bond interface. If the interface belongs to a subnet, you can only change the IP address.

The prefix length can either be specified with the **--address** option, or as a netmask (for IPv4) or prefix length (for IPv6) with the **--netmask** option.

In the following example, an IPv4 address is being associated with interface `ct0.eth1` (both commands are equivalent):

```
purenetwork setattr --address 192.168.0.25/24 ct0.eth1
purenetwork setattr --address 192.168.0.25 --netmask 255.255.255.0 ct0.eth1
```

In the following example, an IPv6 address is being associated with interface `ct0.eth1` (both commands are equivalent):

```
purenetwork setattr --address 2001:0db8:85a3::ae26:8a2e:0370:7334/64 ct0.eth1
purenetwork setattr --address 2001:0db8:85a3::ae26:8a2e:0370:7334 --netmask 64 ct0.eth1
```

Network interface IP addresses and netmasks are set explicitly (DHCP is not supported), along with the corresponding netmasks.

Gateways are specified by IP address. To remove a port's gateway specification, set the **--gateway** option to a null value (`""`). In the following example, a null value is specified to remove the port's gateway specification that was associated with `ct0.eth0`:

```
purenetwork setattr --gateway "" ct0.eth1
```

Change the maximum transmission unit (MTU) of an interface by specifying **--mtu**. If you are changing the MTU of a physical interface that is associated with a VLAN, verify the MTU of the physical interface is greater than or equal to ($\geq$) the MTU of the VLAN interface.

Apps

The Purity Run platform extends array functionality by integrating add-on services into the Purity//FA operating system. Each service that runs on the platform is provided by an app.

For each app that is installed, one app management interface is created per array management interface. An app data interface may also be created for high-speed data transfers.

The naming convention for app interfaces is `APP.datay` for the app data interface, and `APP.mgmty` for the app management interface, where `APP` denotes the app name, and `y` denotes the interface.

The following example lists three app interfaces for the `linux` app. The `linux.mgmt0` interface has been enabled and configured with an IP address to give `pureuser` the ability to log into the app.

```
$ purenetwork list
Name      Enabled Address  ... Speed  Services
...
linux.data0 False   -        ... 10.00 Gb/s app
linux.mgmt0 True    10.8.102.96 ... 1.00 Gb/s app
linux.mgmt1 False   -        ... 1.00 Gb/s app
...
```

For more information about Apps, refer to [pureapp\(1\)](#).

Network Bonding

A bond interface combines two or more similar Ethernet interfaces, such as two or more 1GbE or two or more 10GbE interfaces, to form a single virtual "bonded" interface. A bond interface provides higher data transfer rates, load balancing, and link redundancy.

The **`purenetwork setattr --addsubinterfacelist`** command adds Ethernet interfaces to a bond interface. When you add an Ethernet interface to a bond interface, it becomes a slave to the bond interface. An Ethernet interface can only belong to one bond interface at a time. Administrative slaves cannot be added as slaves to bond interfaces. Furthermore, bond interfaces cannot be slaves to other bond interfaces.

The **`purenetwork setattr --remslavelist`** command removes slaves from a bond interface. When a slave is removed from its bond interface, it becomes an Ethernet interface.

The **`purenetwork delete`** command deletes a bond interface. Before a bond interface can be deleted, it must be empty of all slaves.

Subnets

Interfaces with common attributes can be organized into subnetworks, or subnets, to enhance the efficiency of app, data (iSCSI or NVMe-RoCE), management, and replication traffic.

In Purity//FA, subnets can include physical, virtual, bond, and VLAN interfaces.

Physical, virtual, and bond interfaces can belong to the same subnet. To configure a subnet with physical, virtual, or bond interfaces:

1. Create the subnet. Run **puresubnet create** to create the subnet.
2. Add the physical, virtual, and bond interfaces to the subnet. Run **purenetwork setattr --subnet** to add a physical, virtual, or bond interface to a subnet.

To remove a physical, virtual, or bond interface from a subnet, set the **--subnet** option to a null value.

VLAN interfaces can belong to subnets, but they cannot be mixed with other interface types. All of the VLAN interfaces within a subnet must be in the same VLAN.

To configure a subnet with VLAN interfaces:

1. Create a subnet, assigning a VLAN ID to the subnet. Run **puresubnet create --vlan** to create the subnet.
2. Run **purenetwork create vif --subnet** to create a VLAN interface. The VLAN ID of the VLAN interface must match the VLAN ID of the subnet. Run the command for each corresponding physical network interface to be associated with the VLAN. To remove a VLAN interface from a subnet, run **purenetwork delete**.

For more information about subnets, refer to [puresubnet\(1\)](#).

Subnets and VLAN Interfaces

VLAN tagging allows customers to isolate traffic through multiple virtual local area networks (VLANs), ensuring data routes to and from the appropriate networks.

The **purenetwork create vif** command creates VLAN interfaces.

Create a VLAN interface for each of the corresponding physical network interfaces you want to associate with a VLAN.

In Purity//FA, VLAN interfaces have the naming structure `ctx.ETHy.z`, where *x* denotes the controller (0 or 1), *y* denotes the interface (0 or 1), and *z* denotes the VLAN ID number. For example, `ct0.eth1.500`.

The `purenetwork create vif` command requires the **--subnet** option, which adds the new VLAN interface to the appropriate subnet. Run `puresubnet list` to display a list of all subnets on the array. Run `puresubnet create` to create a new subnet.

All interfaces, including VLAN interfaces, inherit the MTU value of its subnet. Note that the MTU of a VLAN interface cannot exceed the MTU of the corresponding physical interface.

VLAN is only supported for the iSCSI service type, so before creating a VLAN interface, verify the iSCSI service is configured on the physical interface.

When creating a VLAN interface, include the **--address** option to create a reachable VLAN interface.

After a VLAN interface has been created and attached to a subnet, the interface inherits the mask, gateway, MTU, and VLAN ID from the subnet. Likewise, the subnet inherits the service type (for example, iSCSI) from its interfaces.

For more information about subnets and VLAN tagging, refer to [puresubnet\(1\)](#).

Network Ping and Trace

The **purenetwork ping** subcommand is used to determine whether a remote computer can be accessed by the array, provided that the remote computer is ICMP-enabled. The ping target can be specified by IP address. If a DNS service is available and has been configured for the array, the ping target can be specified by hostname.

The **purenetwork trace** subcommand traces and displays the route to a remote computer identified by IP address or, if a DNS service is available and configured for the array, by hostname.

Examples

Example 1

```
purenetwork enable ct0.eth1
```

Enables controller `ct0`'s administrative network interface `eth1` to communicate with the administrative network.

Example 2

```
purenetwork list
```

Lists the attributes of the all network interfaces on the array.

Example 3

```
purenetwork setattr ct0.eth1 --address 192.168.0.24 --netmask 255.255.255.0  
purenetwork setattr ct0.eth1 --address 192.168.0.24/24
```

Assigns IPv4 address `192.168.0.24` to administrative Ethernet interface `ct0.eth1`. Both commands are equivalent.

Example 4

```
purenetwork setattr --addsubinterfacelist eth2,eth3 bond009
```

Adds Ethernet interfaces `eth2` and `eth3` as slaves to bond interface `bond009`.

Example 5

```
purenetwork setattr --address 10.8.108.165 linux.mgmt0
purenetwork enable linux.mgmt0
```

Assigns IP address `10.8.108.165` to app management interface `linux.mgmt0` and enables the interface, giving `pureuser` the ability to log into the `linux` app.

Example 6

```
purenetwork create vif --address 192.168.0.24 --subnet ESXHost001 ct0.eth5.500
```

Creates VLAN interface `ct0.eth5.500` with IP address `192.168.0.24`, and adds it to existing subnet `ESXHost001`. The new VLAN interface inherits the mask, gateway, MTU, and VLAN ID from subnet `ESXHost001`.

Example 7

```
purenetwork delete ct0.eth5.500
```

Removes VLAN interface `ct0.eth5.500` from its subnet and then deletes it.

Example 8

```
purenetwork setattr --subnet "" ct0.eth5
```

Removes physical interface `ct0.eth5` from its subnet. The interface is not deleted.

Example 9

```
purenetwork disable replbond
purenetwork setattr --remslavelist eth2,eth3 --address "" --netmask "" --gateway "" replbond
purenetwork delete replbond
```

Disables the `replbond` bond interface, removes slaves `eth2` and `eth3` from the bond interface and clears the bond interface of all settings, and deletes the `replbond` bond interface.

Example 10

```
purenetwork ping --count 100 myhost.mydomain.com
```

Sends 100 ICMP ping messages to host `myhost.mydomain.com` and displays the response received for each one.

Example 11

```
purenetwork trace 192.168.0.1
```

Displays the route taken by a ping request to network address `192.168.0.1`.

See Also

[pureapp\(1\)](#) [purearray\(1\)](#) [puredns\(1\)](#) [pureds\(1\)](#) [puresubnet\(1\)](#) [purevol-list\(1\)](#)

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